

Let $f : E \rightarrow A \otimes B$ and $g : D \rightarrow E \otimes C$.

$$\begin{array}{ccc}
(A \otimes B) \otimes C & \xrightarrow[\sim]{\alpha_{ABC}} & A \otimes (B \otimes C) \\
\downarrow \text{Hom}(-, D) & & \downarrow \text{Hom}(-, D) \\
\text{Hom}((A \otimes B) \otimes C, D) & \xrightarrow[\text{associator induces isomorphisms under yoneda}]{\alpha_{A,B,C}} & \text{Hom}(A \otimes (B \otimes C), D) \\
\downarrow \text{semisimplicity} \parallel & & \downarrow \text{semisimplicity} \parallel \\
\bigoplus_{c \in \text{Irr}(\text{CAT})} \text{Hom}(A \otimes B, c) \otimes \text{Hom}(c \otimes C, D) & \cong & \bigoplus_{d \in \text{Irr}(\text{Cat})} \text{Hom}(A \otimes d, D) \otimes \text{Hom}(B \otimes C, d) \\
\parallel & & \parallel \\
\bigoplus_c V_c^{AB} \otimes V_D^{cC} & \cong & \bigoplus_c V_D^{Ac} \otimes V_c^{BC} \\
\mathfrak{m} & & \mathfrak{m} \\
|(A, B); c; \mu\rangle \otimes |c, C; D; v\rangle & \xrightarrow[\text{6j-symbols}]{F_D^{ABC}} & \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(c\mu v)}^{(f\alpha\beta)} |A, f; D; \beta\rangle \otimes |B, C; f; \alpha\rangle \\
\mu \in 1, \dots, N_c^{AB} \quad v \in 1, \dots, N^{cC, D} & &
\end{array}$$

$$\begin{array}{ccc}
\begin{array}{c} A \quad B \quad C \\ \diagdown \quad \diagup \quad \diagup \\ \bullet \quad \mu \\ \quad \diagdown \quad \diagup \\ e \quad \bullet \quad v \\ \quad \diagdown \quad \diagup \\ \quad \quad D \end{array} & = & \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(e\mu v)}^{(f\alpha\beta)} \begin{array}{c} A \quad B \quad C \\ \diagdown \quad \diagup \quad \diagup \\ \quad \quad \alpha \\ \beta \quad \bullet \quad f \\ \quad \diagdown \quad \diagup \\ \quad \quad D \end{array}
\end{array}$$

$$\begin{array}{ccc}
(A \otimes B) \otimes C & \xrightarrow[\sim]{\alpha_{ABC}} & A \otimes (B \otimes C) \\
\downarrow \text{Hom}(-, D) & & \downarrow \text{Hom}(-, D) \\
\text{Hom}((A \otimes B) \otimes C, D) & \xrightarrow[\sim]{\alpha_{A,B,C}} & \text{Hom}(A \otimes (B \otimes C), D) \\
\downarrow \text{semisimplicity} \quad \parallel & \text{associator induces isomorphisms under yoneda} & \downarrow \text{semisimplicity} \quad \parallel \\
\bigoplus_{c \in \text{Irr}(\text{CAT})} \text{Hom}(A \otimes B, c) \otimes \text{Hom}(c \otimes C, D) & \cong & \bigoplus_{d \in \text{Irr}(\text{Cat})} \text{Hom}(A \otimes d, D) \otimes \text{Hom}(B \otimes C, d) \\
\parallel & & \parallel \\
\bigoplus_e V_e^{AB} \otimes V_D^{eC} & \cong & \bigoplus_f V_D^{Af} \otimes V_f^{BC} \\
\cap & & \cap \\
|(A, B); e; \mu\rangle \otimes |e, C; D; v\rangle & \xrightarrow[\text{6j-symbols}]{F_D^{ABC}} & \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(e\mu\nu)}^{(f\alpha\beta)} |A, f; D; \beta\rangle \otimes |B, C; f; \alpha\rangle
\end{array}$$

$\mu \in 1, \dots, N_e^{AB}$ $v \in 1, \dots, N_e^{eC, D}$





